

1. Answer the following questions

(a) (16 p.) Suppose the economy described by the Solow model with labour augmenting technological progress operates along the balanced growth path. It is known that the evolution of labour augmenting technological progress is given by $A(t) = 3 * 1.05^t$, depreciation rate is 3%, saving rate is 25% and population growth rate is 4%.

(i) Derive the current growth rate of GDP. Explain carefully.

BGP implies that $Y / AN = \text{const} \Rightarrow$ nominator and denominator must grow at the same rate

$$Y / AN = \text{const} \Rightarrow g_Y = g_{AN} = \frac{3 * 1.05^t * N_0 1.04^t}{3 * 1.05^{t-1} * N_0 1.04^{t-1}} - 1 = 1.05 * 1.04 - 1 = 0.092 \Rightarrow 9.2\% \text{ (8 p.)}$$

The time is discrete, not continuous (it follows from the description of $A(t)$)

(ii) Derive the current growth rate of real profit of the representative firm. Explain carefully.

Profit does not change over time as it equals 0 due to CRS at any t (proof) (8 p.)

As $zF(K, AN)$ is CRS for any A then Euler equation implies: $zF(K, AN) - zF'_N N - zF'_K K = 0$

As markets are perfectly competitive in this model every factor is paid its MP: $w = zF'_N$; $r + \delta = zF'_K$

$$\Pi^{REAL} = zF(K, AN) - wN - (r + \delta)K = zF(K, AN) - zF'_N N - zF'_K K = 0 \text{ (Euler equation)}$$

Thus profit does not grow over time

Some students claimed that profit is zero due to perfect competition with free entry but this argument is incorrect as Solow model assumes one representative firm (no free entry, just one firm performs like a price taker)

(b) (16 p.) Suppose that due to some market frictions at current period equilibrium we observe that: $MU_c = 5$, $MU_l = 3$, $MPN = 2$, where MU stays for the marginal utility and MPN stays for the marginal product of labour.

(i) Explain carefully whether the output produced is too high or too low in comparison with the socially efficient one.

Art current allocation $MRS_{lc} = MU_l / MU_c = 3 / 5 < 2 = MPN$

It means that if the agent sacrifices one (small) unit of leisure then to keep $U = \text{const}$ we need to compensate him with 0.6 units of extra consumption good. At the same time during this (small) unit of time 2 extra units of output will be produced, thus when he gets more than 0.6 units and become better off. Thus we managed to produce a Pareto improvement by producing some extra output, which means that current output is too low (8 p.)

Note:

(1) efficiency has nothing to do with prices

(2) you cannot assume that $MPN = w$ or $MRS_{lc} = w$ as it is stated that this economy has some market frictions

(ii) Provide at least two different frameworks studied in our course that may generate such an equilibrium (i.e. explain the possible reasons). Do not produce graphs.

Marking: (4 p.) per model (not more than 8p in total)

Model 1: money intertemporal model with CIA (not just CIA but the particular model should be indicated)

Explanation:

As h/h is subject to CIA (comments on CIA must be provided) UMP implies $MRS_{lc} = w / (1 + i)$

Firms (are not subject to CIA) and PMP implies $MRS_{lc} = w$

If $i > 0 \Rightarrow MRS_{lc} = w / (1 + i) < w = MP_N$

Comment: symmetric case where firms are subject to CIA might be considered as well

Model 2: Keynesian model where firms are price makers at output market but price takers at labour market in the LR (under flexible prices)

UMP implies $MRS_{lc} = w$

PMP: $w = MRPN^{REAL} = MR * MPN / p = (1 - 1/\epsilon) MPN < MPN$

Thus: $MRS_{lc} = w = MRPN^{REAL} = (1 - 1/\epsilon)MPN < MPN$

Some other examples might be provided but the model should be correctly specified (2-period model) and the condition $MRS_{lc} < MP_N$ must be derived

(c) (16 p.) Consider a forward looking representative agent that lives for three periods ($t=1,2,3$) and his budget constraint

takes the form $c_1 + \frac{c_2}{1+r} + \frac{c_3}{(1+r)^2} = y_1^d + \frac{y_2^d}{1+r} + \frac{y_3^d}{(1+r)^2}$, where y_i^d stays for the real disposable income in period i .

Government planning horizon is the same (three periods) and as government can borrow and lend at the same interest rate as the representative agent. Assume that disposable incomes in the three periods are given by: 13, 110 and 242 and real interest rate is 10%.

(i) State a verbal definition of permanent income and derive the permanent income for this agent.

Verbal definition [it must be a rigorous def, i.e. following this definition it should be possible to get the correct numerical answer for the Permanent income] **(3 p.)**

Derivation: $y^p \left[1 + \frac{1}{1.1} + \frac{1}{1.1^2} \right] = 13 + \frac{110}{1.1} + \frac{242}{1.1^2} = 113 + 200 = 313$

$$y^p = \frac{313}{1 + \frac{1}{1.1} + \frac{1}{1.1^2}} = \frac{313 \times 121}{331} \approx 0.95 \times 121 \approx 114 \text{ (5 p.)}$$

Comment: this task did not ask about PIH, it asked about PI only!

(ii) Suppose that before the start of period 2 the government unexpectedly provides a transfer equal to 22 that is financed by issuing government bonds. Do you expect this agent to spend this transfer (fully or partially)? Explain carefully.

Not spend but fully saved due to RE (proof) **(8 p.)**

Analysis of GBC and combined IBC $\Rightarrow$ the same BS + the same preferences $\Rightarrow$ the same UMP implies the same solution

Thus, no change in consumption, transfer will be saved to finance consumption under increased next period taxes

This is the essence of RE (state the RE)

(d) (16 p.) Consider a Diamond-Dybvig model discussed at the lecture. Suppose that to make banks less vulnerable to runs, bank regulators propose a cap on the amount of interest banks can pay to short-term depositors who withdraw in period 1.

(i) Explain carefully what is the maximum period 1 payout d_1 to depositors for which no bank runs should occur.

As we know in DD considered at the lecture there are 2 NE, one of which includes a bank run

Here you must look at the unpleasant NE (with run) only $\Rightarrow$ no points for the analysis of good equilibrium (with risk sharing)

Max cap = 1 (proof) **(8 p.)**

If $N-1$ agents come at $t=1$ then the last one has no incentive to come if $(1+R)(N-(N-1)d_1) \geq d_1$ (explanation) $\Rightarrow$

$$d_1 \leq \frac{1+R}{1+R-R/N}$$

This condition is satisfied for any $R>0$ and N if we set a cap at $d_1=1$ (actually it can be set a bit higher if we follow this exact formula)

(ii) What would be a disadvantage of capping d_1 at that level?

If $d_1=1$ then $d_2=1+R$

We get the same outcome as in the autarky w/o banks $\Rightarrow$

This cap eliminates the gains from maturity transformation performed by banks (no risk sharing)

As agents are risk averse this reduces their EU while $E\pi_{banks}=0$ anyway $\Rightarrow$ Social welfare is reduced (Pareto Inefficient allocation) **(8 p.)**

Some students claimed that banks profits are reduced but this is just impossible as $E\pi=0$ due to free entry.

2. (36 p.) Consider a small open economy described by the real dynamic model with initial capital stock equal 64. Currently all financial capital flows are completely blocked, and economy is in equilibrium. Current real wage rate is 2 and currently the firm finds profitable to have the same employment in current and future periods. It also plans to increase the capital stock in the future period by 36 units. Assume that depreciation rate is 0. Production function is given by $2\sqrt{KN}$ and total factor productivity stays constant. Assume that world real interest rate is 25%.

(a) Find algebraically the output level and the interest rate. Explain carefully.

Derivation of Y with comments **(6 p.)**

Equilibrium at labour market guarantees that $N_d = N_s$

N_d comes from 2-period PMP and inverse N_d is given by $MPN = W/P \Rightarrow MPN = 8 / \sqrt{N} = 2 \Rightarrow N = 16$

$$Y = 2\sqrt{KN} = 2\sqrt{64 \cdot 16} = 64$$

Derivation of r with comments **(6 p.)**

Initially capital flows are fully blocked, thus domestic interest rate is not tight to the world one.

The PMP provides desired stock of capital for the next period that should satisfy the equation:

$$r + \delta = MPK_{+1}$$

We know that depreciation is absent and $K_{+1} = 64 + 36 = 100$ while employment stays constant $N_{+1} = N = 16$

$$\text{Thus, } r = MPK_{+1} = \sqrt{N_{+1} / K_{+1}} = \sqrt{16 / 100} = 0.4$$

(b) Suppose that government decided to remove any restrictions on capital movements. How this reform would affect current period output, real wage rate and real exchange rate if domestic and foreign goods are homogenous? Show graphically (using graphs for labour and output markets) illustrating initial equilibrium by a black colour (denote it by A) and the new equilibrium by a blue colour (denote it by C). Explain every step. You must start every step with a new paragraph.

Two graphs with initial equilibrium (A) illustrated by black colour, new equilibrium (C) illustrated by blue colour with comments and final answers on ΔY , Δw , ΔRER **(12 p.)**

Note 1: this is a real model (no money, goods and assets are exchanged directly) $\Rightarrow$ no need to know anything about the exchange rate mechanism (fixed or flexible) as adjustment deals with RER, not nominal

Solutions that somehow deal with nominal exchange rates are incorrect

Note 2: r does not change just because you want it to change, r changes in response to ED/ES at goods market. Similarly, real wage changes in response to ED/ES at goods market

Note 3: PPP is not an assumption, this is an implication of goods homogeneity as o/w (if P is different from EP^*) only least expensive good will be demanded and one of the economies will produce nothing

(c) How the result of (b) would change if domestic and foreign goods are heterogeneous? Use your graph from (b) but add a new equilibrium by dotted lines (denote it by D) at both graphs.). Explain every step. You must start every step with a new paragraph.

New equilibrium (D) illustrated by dotted lines at the diagram from (b), not a new diagram!

with comments and final answers on ΔY , Δw , ΔRER **(12 p.)**